

Proposal for Large-Scale Field Testing and Validation of [Project Name / Solution]

Submitted to:

AngloGold Ashanti Iduapriem Mine Management, Tarkwa

Submitted by:

UMaT Students Research Group

University of Mines and Technology (UMaT), Tarkwa

Date: [Insert Date]

1.0 Executive Summary

Brief Problem Statement:

This proposal addresses a key operational challenge faced by AngloGold Ashanti Iduapriem Mine — [insert concise summary of the problem, e.g., “limited data efficiency in geological logging operations,” “delays in field data synchronization,” or “inefficient manual image labeling during core logging”].

Proposed Solution:

The UMaT research team has developed an innovative software suite designed to modernize and streamline geological data management. The system integrates multi-platform access (desktop, web, and mobile) with AI-assisted image recognition to enhance speed, accuracy, and accessibility across departments.

Objective of this Proposal:

To seek approval and support for conducting a **large-scale field test** of the proposed system within a real mining environment to validate its operational efficiency, scalability, and economic value.

Expected Benefits:

- Improved productivity and data accuracy across field teams.
- Time and cost savings through automated image classification and faster data upload.

- Enhanced mobility and safety by reducing reliance on bulky desktop-only systems.
- Strengthened collaboration between AngloGold Ashanti and UMaT.

Estimated Budget & Timeline:

The proposed pilot field test will run for approximately [insert duration, e.g., 6 weeks], at a total estimated cost of ₵[insert figure, e.g., 45,000–75,000 GHS].

2.0 Introduction and Background

2.1 Problem Context

AngloGold Ashanti Iduapriem Mine currently relies on legacy geological data systems (e.g., Fusion Suite–based DH Logger), which have proven to be slow, desktop-restricted, and cumbersome for field operations. These limitations hinder productivity and increase the difficulty of on-site data capture.

2.2 UMaT's Initial Solution

The UMaT research team has designed and prototyped a more adaptive and intelligent digital logging system capable of functioning across all platforms — desktop, tablet, and mobile. Early laboratory tests have demonstrated significant improvements in speed, usability, and data handling efficiency.

2.3 Rationale for Large-Scale Field Testing

The controlled prototype testing has confirmed technical feasibility. The next step is to validate the solution under **real operational conditions**, accounting for variable connectivity, environmental factors, and live workflow integration within the mine's geological processes.

3.0 Project Objectives and Scope

3.1 Primary Objective

To conduct a comprehensive field test that validates the performance, scalability, and operational impact of the new geological data management suite.

3.2 Secondary Objectives

- Assess long-term stability and maintenance requirements.
- Measure productivity and accuracy improvements compared to existing systems.
- Collect quantitative data for a full techno-economic assessment.

3.3 Scope of Work

Inclusions:

- Field testing, data collection, AI image processing validation, and usability evaluation.
- Training sessions for field staff and supervisors.

Exclusions:

- Permanent infrastructure installation.
 - Commercial licensing and full-scale rollout (to be covered in a subsequent proposal).
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4.0 Technical Methodology for Field Testing

4.1 Description of the Technology / Solution

The system integrates a cloud-enabled database, AI image recognition, and multi-platform interfaces to digitize, tag, and manage geological logging data in real time.

4.2 Test Site Selection

The field test will be conducted at **[specific mine section / department]**, selected for its representative geology, accessibility, and operational activity level.

4.3 Experimental Design and Setup

- **Baseline Data Collection:** Record operational benchmarks from the existing DH Logger system.
- **Installation Plan:** Deploy mobile and web interfaces for selected users.
- **Monitoring & Data Acquisition:** Continuous data capture and system performance tracking.
- **Control Parameters:** Standardize data formats, devices, and logging conditions.

4.4 Data Analysis Plan

Data will be analyzed statistically to assess efficiency gains, error reduction, and time savings. Key metrics will include average processing time per dataset, system uptime, and user satisfaction.

5.0 Project Management Plan

5.1 Project Team and Roles

UMaT Team:

List of team members, their responsibilities (e.g., AI development, database management, UI design), and the supervising faculty members.

Mine Personnel:

Mine-side support roles including Section Engineer, Safety Officer, and designated liaison for on-site coordination.

5.2 Timeline and Milestones

Phase	Description	Duration
Phase 1	Pre-Mobilization (Planning, HIRA, Procurement)	1 week
Phase 2	Mobilization & Installation	1 week
Phase 3	Field Testing & Data Collection	3 weeks
Phase 4	Demobilization & Data Analysis	1 week
Phase 5	Reporting & Presentation	1 week

5.3 Risk Assessment and Mitigation

Potential Risk	Mitigation Strategy
Equipment failure	Maintain backup devices and power supply

Potential Risk	Mitigation Strategy
Data loss	Real-time backups and cloud synchronization
Safety incidents	Compliance with mine HSE standards
Delays / weather	Adjust testing windows accordingly

6.0 Health, Safety, and Environmental (HSE) Considerations

- Conduct full **Hazard Identification and Risk Assessment (HIRA)** prior to deployment.
- Adhere strictly to **AngloGold Ashanti's safety protocols**, including Permit-to-Work systems, LOTO, and PPE compliance.
- Ensure environmentally responsible operations — proper disposal of materials, prevention of spills, and minimal ecological disturbance.

7.0 Deliverables

- **Final Technical Report** with detailed analysis and results.
- **Techno-Economic Analysis Report** estimating ROI, cost savings, and productivity improvements.
- **Formal Presentation** of outcomes to mine management.
- All **raw and processed datasets** in approved electronic formats.

8.0 Resource Requirements and Budget

8.1 Personnel

UMaT student researchers, academic supervisors, and potential field assistants.
Modest **stipends and honoraria** are requested to support time, travel, and logistics.

8.2 Materials & Equipment

Item	Source	Purpose	Estimated Cost (GHS)
AI Training Workstation / Cloud Credits	UMaT / Sponsor	Model training and performance testing	10,000 – 15,000
Field Devices (Tablets, Laptops)	UMaT	Data logging and field testing	In-kind
Consumables & Cables	AngloGold	Field setup and connectivity	1,200
Miscellaneous (Transport, fittings, power)	Shared	Logistics and site support	800

8.3 Services

Service	Provider	Estimated Cost (GHS)
Transportation	AngloGold	In-kind
Workshop & Utilities	AngloGold	In-kind
External Lab / Verification	UMaT	800

8.4 Budget Summary

Category	Estimated Cost (GHS)
Personnel (Stipends & Honoraria)	25,000 – 40,000
Materials & Equipment	10,000 – 15,000
Services	800 – 1,500
Total Estimated Cost	¢40,000 – ¢75,000

9.0 Benefits to AngloGold Ashanti Iduapriem Mine

- **Operational Efficiency:** Faster data processing, reduced manual work.
 - **Financial Savings:** Lower overhead and improved decision turnaround.
 - **Safety & Environmental:** Reduced site downtime and digital record-keeping.
 - **Strategic Impact:** Strengthens partnership with UMaT and supports CSR objectives through academic–industry collaboration.
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10.0 Conclusion

This proposal outlines a feasible, low-risk opportunity to pilot an innovative data management solution developed by UMaT.

The project offers measurable operational value, aligns with AngloGold Ashanti’s digital transformation objectives, and provides a model for future collaborations.

The UMaT team respectfully requests **approval and logistical support** to proceed with the proposed large-scale field test.

11.0 Appendices

Appendix A: CVs of Student Researchers and Supervisors

Appendix B: System Architecture and Technical Schematics

Appendix C: Preliminary HIRA Document

Appendix D: Detailed Budget Breakdown

Appendix E: Letters of Support from UMaT Faculty